

BT101

Introductory Biology

(Lecture 3)

Dr. Rajkumar P Thummer
Assistant Professor
Department of Biosciences and Bioengineering
IIT Guwahati
Guwahati

Introduction



Evolution of life:

- ☐ **Origin of Life**
- ☐ Darwin's concepts of evolution
- ☐ Biodiversity

Cell, the structural and functional unit of life:

- ☐ Three domains of life
- ☐ Cell types, cell organelles and structure
- ☐ Basic biomolecules of cell

Introduction



- ❑ Organisms are divided into **two** main classes, initially defined by whether they contain a nucleus.
- ❑ Prokaryotes (bacteria) lack a nuclear envelope; eukaryotic cells have a nucleus in which the genetic material is separated from the cytoplasm.
- ❑ Prokaryotic cells are generally smaller and simpler than eukaryotic cells; in addition to the absence of a nucleus, their genomes are less complex and they do not contain cytoplasmic organelles or a cytoskeleton.

Introduction



Table 1.1 Prokaryotic and Eukaryotic Cells

Characteristic	Prokaryote	Eukaryote
Nucleus	Absent	Present
Diameter of a typical cell	$\approx 1\mu\text{m}$	$10\text{--}100\mu\text{m}$
Cytoskeleton	Absent	Present
Cytoplasmic organelles	Absent	Present
<u>DNA</u> content (base pairs)	1×10^6 to 5×10^6	1.5×10^7 to 5×10^9
Chromosomes	Single circular <u>DNA</u> molecule	Multiple linear <u>DNA</u> molecules

<https://www.youtube.com/watch?v=zZtcMBTQaS4>
<https://www.youtube.com/watch?v=Pxujitlv8wc>
<https://www.youtube.com/watch?v=q9fthbK08Uc>

Introduction

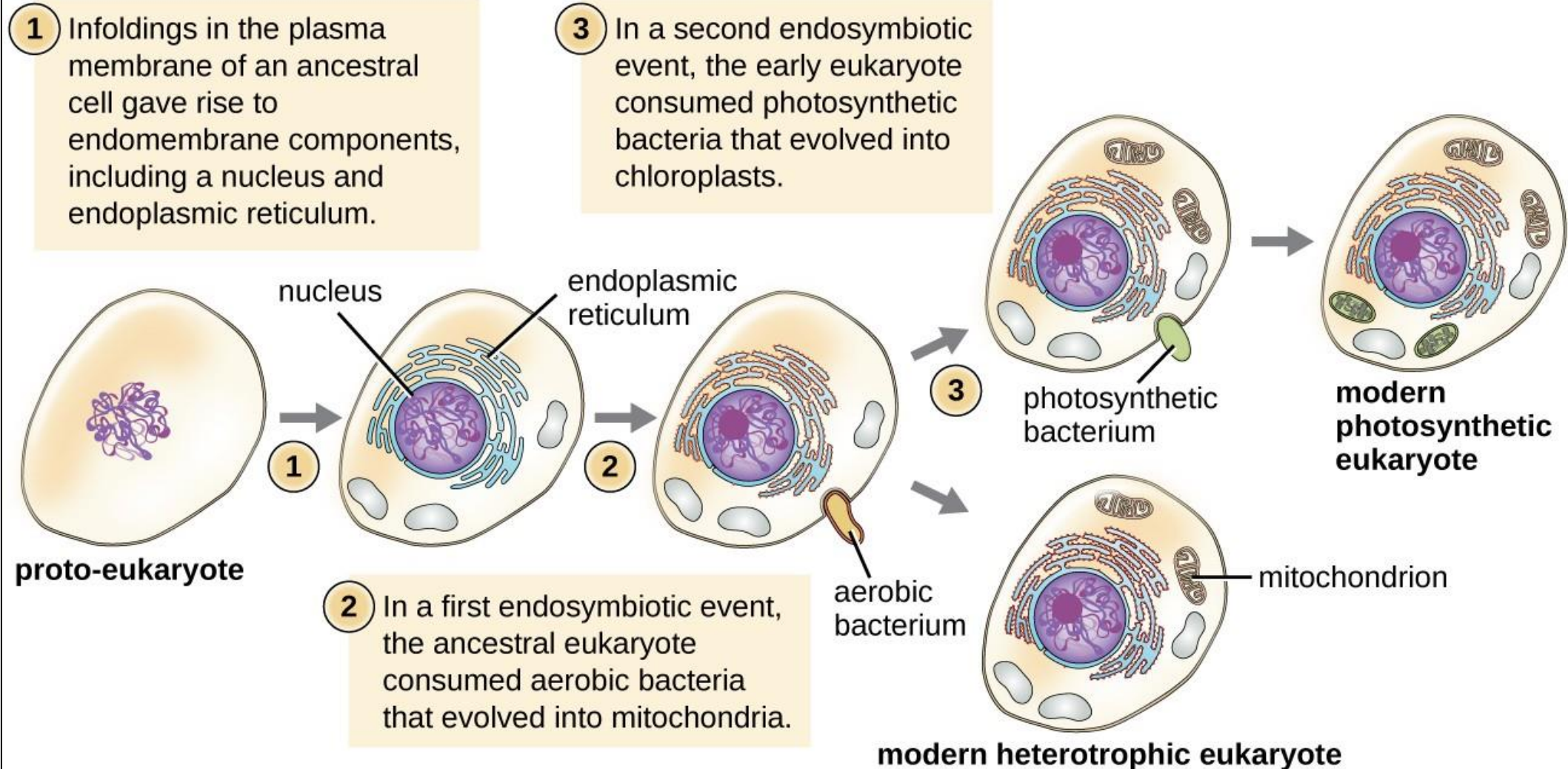


- ❑ **REPRODUCTION:** Asexual (binary fission) vs Sexual Reproduction
- ❑ **RIBOSOMES** (central dogma of molecular biology)
- ❑ **CHROMOSOMES** (circular vs linear; single vs multiple)
- ❑ **Extrachromosomal material** (plasmid)
- ❑ **MITOCHONDRIA vs CHLOROPLASTS**
- ❑ **EXAMPLES** of Prokaryotes and Eukaryotes
- ❑ **EXAMPLES** of Unicellular Eukaryotes

<https://www.youtube.com/watch?v=zZtcMBTQaS4>
<https://www.youtube.com/watch?v=Pxujitlv8wc>
<https://www.youtube.com/watch?v=q9fthbK08Uc>

Endosymbiotic theory

The Endosymbiotic Theory



Evidence of Endosymbiotic theory



- ❑ Presence of circular DNA (similar genomes to prokaryotes)
- ❑ Presence of 70S ribosomes
- ❑ Can self-replicate (through binary fission)
- ❑ Size (around 10 μm)
- ❑ Have two membranes
- ❑ Porins and cardiolipin are only found in bacteria, mitochondria and chloroplasts
- ❑ Location of Electron Transport Chain

Important Links (Endosymbiotic theory)



- ❑ <https://www.youtube.com/watch?v=8oSqXAwLsZc&t=158s>
- ❑ <https://www.youtube.com/watch?v=eCWwKrJyKxA>
- ❑ <https://www.youtube.com/watch?v=FGnS-Xk0ZqU>
- ❑ <https://www.youtube.com/watch?v=bBjD4A7R2xU>
- ❑ <https://www.youtube.com/watch?v=4LhBZ2H5SwM>
- ❑ <https://www.youtube.com/watch?v=Fdh62URT5xE>

Thank you for your attention